

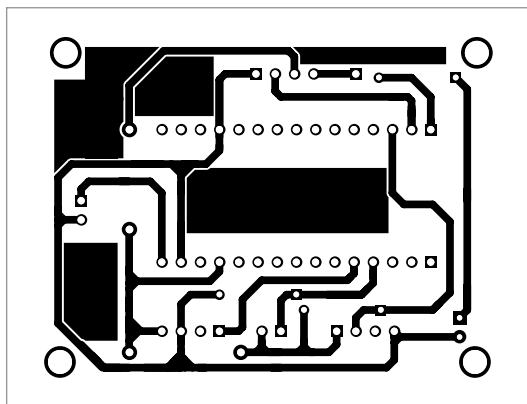


Fig. 4: Actual-size PCB layout for sensor node

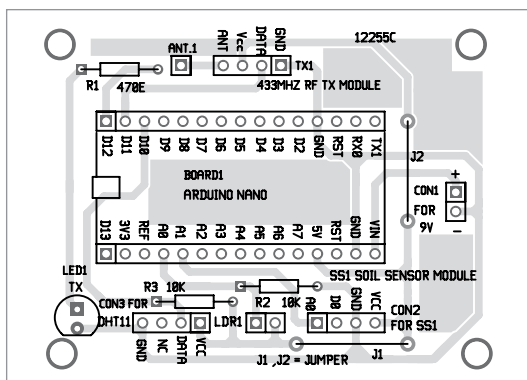


Fig. 5: Components layout of the PCB in Fig. 4

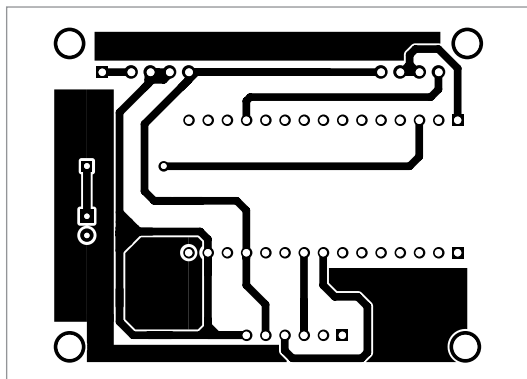


Fig. 6: Actual-size PCB layout for receiver

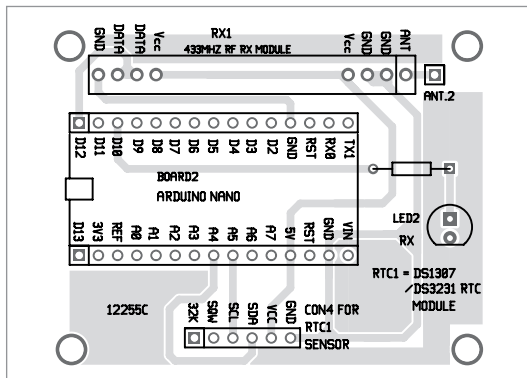


Fig. 7: Components layout of the PCB in Fig. 6

es. So, the analogue output voltage from LDR1 is directly proportional to ambient light. Arduino converts this analogue voltage into digital value (between 0 and 1023) and maps it between 0 and 99%. Finally, it converts this percentage value into ASCII characters.

Soil moisture sensor (SS1) produces analogue voltage output inversely proportional to soil moisture content. That means its output voltage decreases as soil moisture content increases. Arduino converts this analogue voltage into digital value (between 0 and 1023) and maps it between 0 and 99%. Finally, it converts this percentage value into ASCII characters.

And then these data are sent to RF transmitter module (TX1) through pin D11 of the Arduino.

TX1 gets this string of ten characters (we may call it as packet), modulates it using 433MHz carrier and transmits it through antenna ANT1. Every time the packet is transmitted, the microcontroller blinks the LED1 to indicate data is being transmitted. This cycle repeats continuously after every set time period (say, 10 seconds).

Remote data receiver

As shown in the circuit diagram of remote data receiver in Fig. 3, there are only three major components in this receiver circuit:

1. 433 MHz RF receiver module RX1 that has four interfacing pins: Vcc, Gnd, Data Out, and Antenna. Vcc pin is connected to 5V output of Arduino board and Gnd is connected to common ground. Data pins are connected to digital pin D12 of Arduino board. An antenna (ANT2), same as transmitter side, is con-

nected to antenna pin.

2. RTC module used here is a DS1307 RTC chip having four interfacing pins: Vcc, Gnd, SDA, and SCL. Vcc pin is connected to 5V output of Arduino board and Gnd is connected to common ground. SDA and SCL pins are for TWI (IIC) byte oriented data transfer. These are connected to A4 (SDA) and A5 (SCL) pins of Arduino Nano board, respectively.

3. LED (LED2), which is connected to digital pin D10 of Arduino through current-limiting resistor R4.

The Arduino Nano board and RF RX1 module get power supply from central storage system computer through USB cable. The Arduino Nano board also logs data and communicates with computer using the same USB cable.

Working and operation of the remote data receiver can be understood from the following steps:

- The receiver first checks whether RTC is running and giving correct date and time or not. If RTC is running, Arduino displays current time on the serial monitor.

- The RF RX1 module receives packets from every node. It demodulates and gives these packets to Arduino microcontroller.

- The microcontroller gets the packets and extracts all four values from each packet: ambient light, soil moisture content, temperature, and humidity along with node number.

- It serially sends following data to computer through USB as Node x:

- Soil moisture: XX %
- Light: XX %
- Humidity : XX %
- Temperature : XX °C

- The microcontroller blinks LED2 to indicate data is being received.

- Also, each time Arduino gets data from any sensor node, it reads time and date from RTC module and serially sends it to computer. So, the sensor node data values are printed along with time and date.

- All four sensor data values with time and date are displayed on serial monitor in sequence.

- This process repeats for each packet received from any node at any time. As all nodes are synchronised in time, they transmit data sequentially one after another.